Aspandiyar Kurman

Creative Technologist — Game Developer

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Profile

3D artist at Digiteq Automotive's Virtual Reality Lab since 2023: optimised game assets for VR simulations. Builds real-time UE5 systems in C++, HLSL and Python: a gamepad-played VJ engine, a 60-dancer crowd with an autonomous camera, GPU simulations and frame-time self-tests. Code written with AI coding agents under my direction.

Experience

3D Artist — Digiteq Automotive, Virtual Reality Lab 2023 – now

- Game-asset creation: modelling, texturing, optimisation
- Preparing optimised assets for 3D simulations in VR

Independent real-time & audio-reactive work — Self-directed ongoing

- Unreal Engine 5 live VJ engine in C++ and HLSL, played with a gamepad
- Houdini, Blender and UE5 pipelines driven from Python
- TouchDesigner GLSL visualizers with a phone web control panel
- Concept, system design, art direction and testing; code written with AI coding agents under my direction

3D Artist (internship) — Cinesco Jun – Sep 2023
Game-asset design, modelling and sculpting

HMI Front-end Developer — Digiteq Automotive, DTF-D 2023
HMI skinning; UI installation for Škoda, debugging and testing

Market Researcher — Copy General Technologies 2022
Research of the Czech market and 3D-print technologies; 3D modelling, consulting, customer service

Printing Operator, junior → senior — Copy General 2019 – 2022
Typography design services and their sale; order fulfilment, email and phone communication

Selected projects

- **AudioVJ** *(ongoing)* — Standalone UE5 VJ app played live on a gamepad; a custom audio analyzer drives six real-time scenes, scripted self-tests check frame time. UE5, C++, HLSL, Python · Case study
- **Banana Party in Heaven** — 60 cartoon bananas dance on the beat through a blueprint-free C++ AnimInstance, filmed by an autonomous camera director. UE5, C++, HLSL · Case study
- **Liquid / Obsidian Bloom** — Four real-time liquids at 2.2–2.4 ms GPU time per frame (1080p), gated by a scripted test for GPU time, beat timing and NaNs. UE5, C++, HLSL · Case study
- **Triad** *(ongoing)* — ~2,000-line UE 5.8 C++ plugin decodes OSC on its own UDP thread; 250,126 Houdini Vellum grains play back in Unreal as GPU instances. UE5, C++, HLSL, Houdini · Case study

Skills

Engines: Unreal Engine 5 (C++, Python editor scripting, DLSS 4.5), Unity, TouchDesigner
Code & shaders: C++, HLSL, GLSL, Python, VEX, JavaScript
Real-time: frame-time budgets, GPU particle and liquid simulation, deterministic fixed-step recording, automated self-tests
3D & art: Blender (modelling, sculpting, bpy automation, Geometry Nodes, Rigify), Substance 3D Painter, Houdini (Vellum, Karma), Photoshop
AI-assisted development: directing AI coding agents that write the code while owning concept, system design, testing and review; Tripo and Kimodo for assets and motion
Pipeline: FFmpeg, PowerShell, Python build steps (Unreal materials generated from HLSL, headless Blender scene builds)
Soft skills: teamwork, communication, creativity, management, presentation, organisation

Education

Secondary School of Glassmaking — Kamenický Šenov · glass cutting and engraving; design concepts 2018 – 2019